
Reasoning and Decision Making under Uncertainty

Stochastic Processes

Prof. Dr. Frank Deinzer

Technical University of Applied Sciences Würzburg-Schweinfurt
Faculty Computer Science and Business Information Systems

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Content and Overview

► Stochastic Processes

► Statistical Classification

- Static, no time-dependent classification
- Bayes classifier

► Context Classification

- Time-dependent classification
- State estimation
- Hidden Markov Models
- Recursive Density Estimation

Content and Overview

► Stochastic Processes

- Important univariate densities
 - Discrete and continuous densities
 - Different types of distributions
 - Univariate and multivariate densities
 - Sampling from distributions
 - Estimating distribution parameters
- Multivariate densities
- Importance sampling
- Mixture models
 - EM algorithm
- Measures for similarity of densities
 - Kullback–Leibler divergence

Stochastic Processes

- ▶ Stochastic process: mathematical description of random processes
 - What do we know about the random process of throwing a dice?
 - Number of points on dice is a **random variable**
 - Structure of stochastic processes
 - Discrete random variable $\Leftrightarrow$ Continuous random variable
 - One random variable $\Leftrightarrow$ Multiple random variables (Unimodal $\Leftrightarrow$ Multimodal)
 - Underlying distribution describes stochastic process
 - Uniform distribution
 - Multinomial distribution
 - Normal distribution
 - Exponential distribution
 - Mixture distribution
 - We will address the following questions
 - Create random numbers that follow a distribution $\rightarrow$ sample from distribution
 - Estimate parameters of distributions from samples of given sample set



Stochastic Processes – Uniform Random Numbers

► Creating random numbers

- One often needs to create random numbers that follow a specific distribution
- We refer to this as *sampling from a distribution*

- Sampling from discrete **unimodal uniform distribution**

- Create uniform pseudo-random numbers: linear congruential generator

$$Z_{n+1} = (aZ_n + b) \bmod m$$

- Need the following components for algorithmic implementation

- Multiplier a
- Increment b
- Modulus m
- Start value Z_0 (called seed)

	m	a	b	Output (Bits)
Java: java.util.Random	2^{48}	25214903917	11	47 ... 16
Numerical Recipes	2^{32}	1664525	1013904223	alle
glibc (GCC)	2^{31}	1103515245	12345	30...0
Microsoft Visual C++	2^{32}	214013	2531011	30...16

- How to sample from distribution of dice?

Stochastic Processes – Important Unimodal Densities

- ▶ Random variable x that follows distribution defined by parameters a

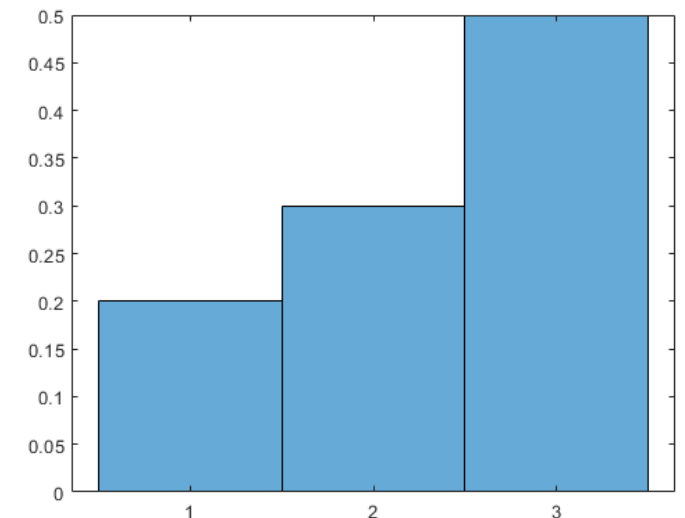
$$p(x|a)$$

- The parameters describe all possible distributions in a very general way

- ▶ Important distributions

- Most basic: discrete uniform distribution
 - Exactly what a throwing a dice or flipping a coin does
 - All possible results have same probability
- Important: discrete multinomial distribution
 - Urn problem: Random drawing of colored balls (with putting back) from an urn
 - Discrete and finite possible values for random variable
 - Each value has individual probability of occurrence

```
data= [ 2 3 3 2 1 3 2 3 3 3 3  
3 2 2 3 1 1 1 3 1 2 3 3 2 3 1  
2 2 3 2 3 2 1 1 3 3 3 3 3 2 ]
```

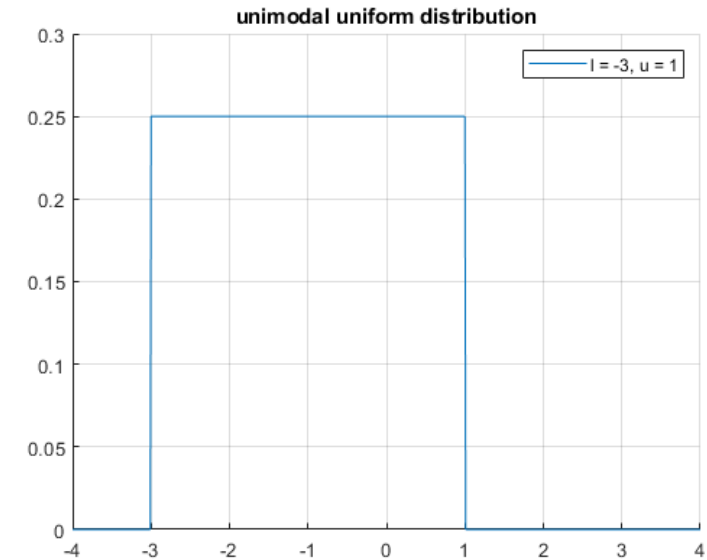


Stochastic Processes – Important Unimodal Densities

- Most important: continuous unimodal uniform distribution
 - Described by parameters lower and upper limit

$$p(x|\mathbf{a}) = p(x|l, u) = \begin{cases} \frac{1}{u-l} & l \leq x \leq u, \\ 0 & x < l \text{ or } x > u \end{cases}$$

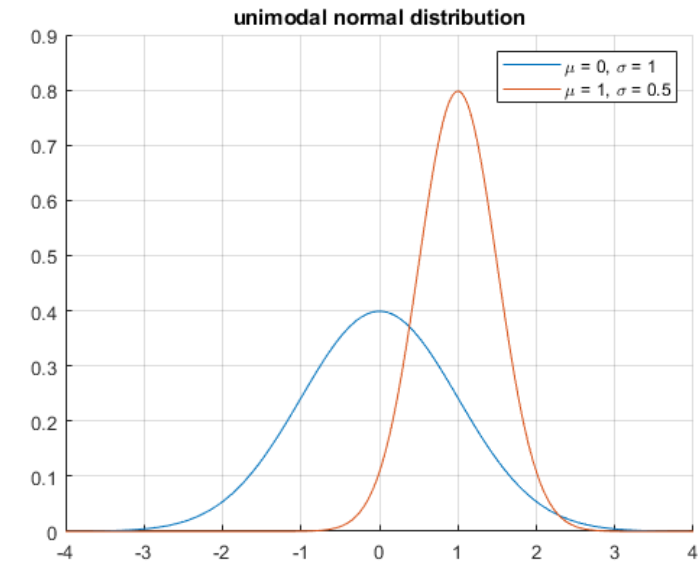
- All numbers from range have same probability



- Most famous: unimodal normal distribution (Gaussian)
 - Described by parameters mean μ and standard deviation σ

$$p(x|\mathbf{a}) = p(x|\mu, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

- Mean is the value where the distribution has maximum value



Stochastic Processes – Important Unimodal Densities

- Unimodal exponential distribution

$$p(x|\mathbf{a}) = p(x|\lambda) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

- Usage: Generate statistically motivated density values from distance measurements

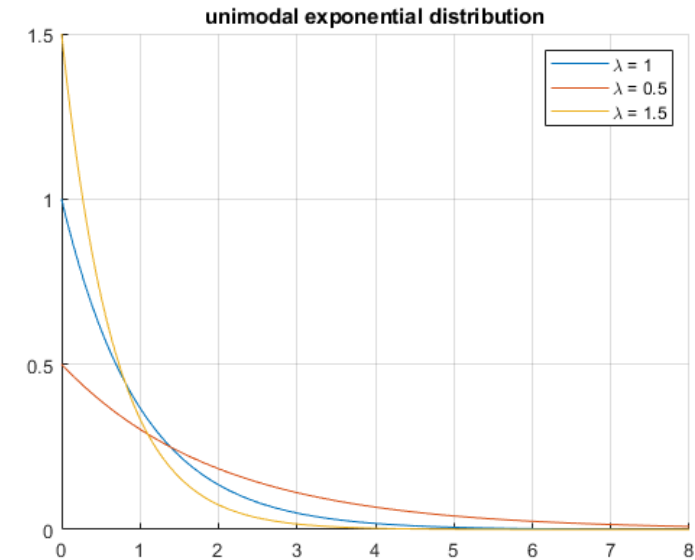
- There are many more distributions

- Poisson distribution, von Mises distribution, Binomial distribution, Erlang distribution, several skewed distributions, ...

- Cumulative distribution functions

- Probability that random variable x will take a value less than or equal to a constant X

$$P(x < X|\mathbf{a}) = \int_{-\infty}^X p(x|\mathbf{a}) dx$$



Stochastic Processes – Sampling Unimodal Densities

- ▶ Creating random numbers (sampling from distributions)
 - Random number $U(0, 1)$ in range $[0.0, 1.0]$
 - Create uniform pseudo-random numbers from linear congruential generator
 - Divide by largest possible value
 - Sampling from standard-normally distributed distribution (mean 0, standard deviation 1.0)
 - Generate two independent, uniformly distributed random numbers

$$u_1, u_2 \sim U(0, 1)$$

- Box-Muller method: calculate two independent, standard-normal distributed random numbers

$$z_1 = \cos(2\pi u_1) \sqrt{-2 \ln u_2}$$

$$z_2 = \sin(2\pi u_1) \sqrt{-2 \ln u_2}$$

$$z_1, z_2 \sim \mathcal{N}(0.0, 1.0)$$

Stochastic Processes – Sampling Unimodal Densities

- Sampling from an arbitrary distribution
 - Inverse transform sampling: generate other probability distributions from uniformly distributed random numbers
 - Given a density function p
 - Calculate the cumulative distribution function P and its inverse P^{-1} (aka quantile function)
 - Generate an independent, uniformly distributed random number u
 - Use this random number to calculate a random number z that follows p

$$u \sim U(0, 1)$$

$$z = P^{-1}(u)$$

Stochastic Processes – Sampling Unimodal Densities

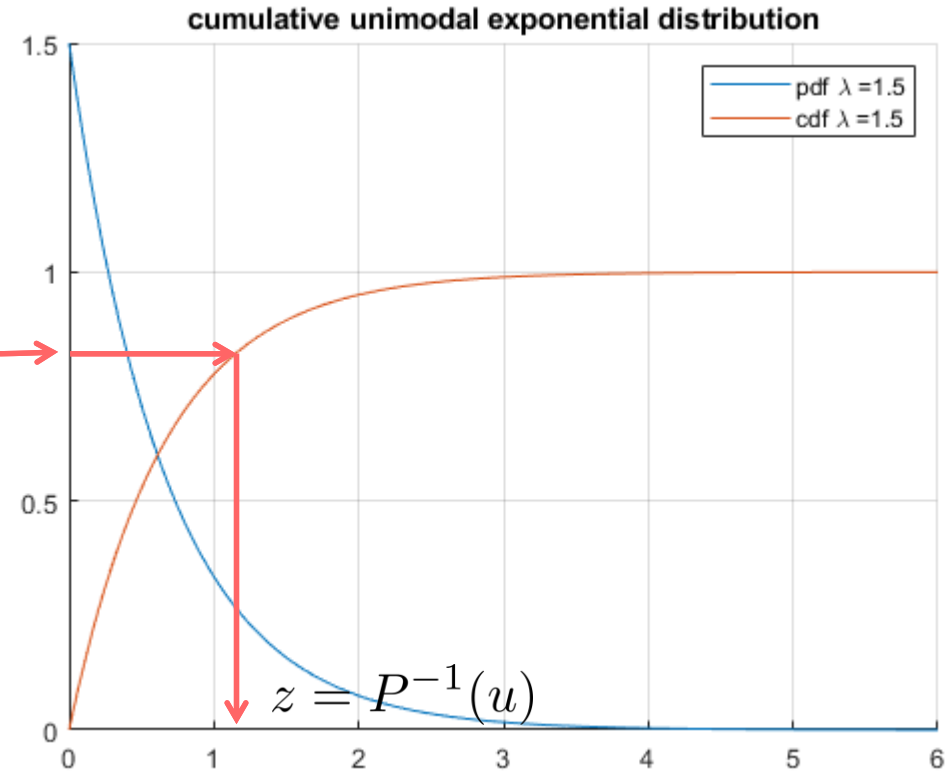
- Example: sample from exponential distribution, create exponentially distributed random number

- Create uniform random number

$$u \sim U(0, 1)$$

- Evaluate inverse of cumulative distribution function:

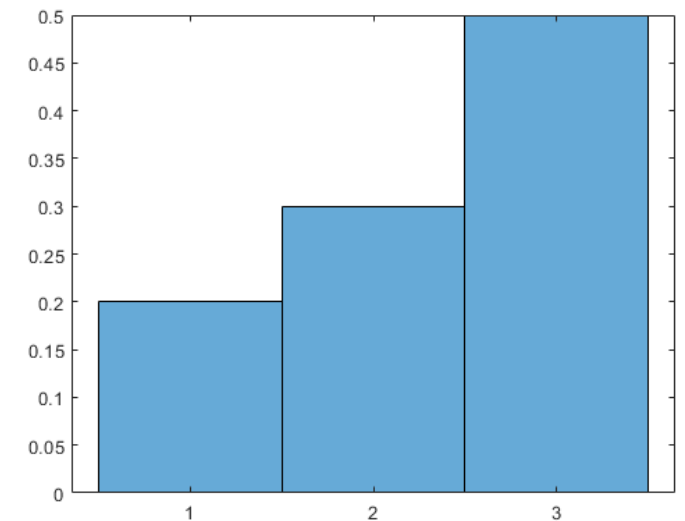
Here: graphical solution for illustration



Stochastic Processes – Sampling Unimodal Densities

- Sampling from multinomial distribution
 - Goal: Draw possible elements proportional to their probability
 - Algorithm:
 - Create array of cumulative probabilities/ranges
 - Create uniform random number $U(0, 1)$
 - Find the element in the array whose cumulative range contains the random number
 - Computational complexity of operation
 - Linear search: $O(n)$
 - Binary search: $O(\log n)$

```
data= [ 2 3 3 2 1 3 2 3  
3 3 3 3 2 2 3 1 1 1 3 1  
2 3 3 2 3 1 2 2 3 2 3 2  
1 1 3 3 3 3 3 2 ]
```



Element	Probability	Cumulative Range
1	0.2	0.0 – 0.2
2	0.3	0.2 – 0.5
3	0.5	0.5 – 1.0

Stochastic Processes – Parameter Estimation

► Estimation of distribution parameters

- Density parameters are usually unknown for concrete problems
 - Must be estimated with the help of a set of samples ω , e.g.

$$\omega = \{3, 1, 9, 6, 4, 3, 7, 2, 4, 5, 7\}$$

- Estimation of parameters of unimodal uniform distribution
 - Necessary parameters: lower and upper boundaries

$$p(x|l, u) = \begin{cases} \frac{1}{u-l} & l \leq x \leq u, \\ 0 & x < l \text{ or } x > u \end{cases}$$

- Identify the maximum likelihood (ML) estimators for lower and upper boundaries
- Find minimum and maximum values in sample set

$$l = \min \{x | x \in \omega\}$$

$$u = \max \{x | x \in \omega\}$$

Stochastic Processes – Parameter Estimation

- Estimation of unimodal normal distribution
 - Necessary parameters: mean and standard deviation

$$p(x|\mu, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

- Maximum likelihood (ML) estimator for mean and variance resp. standard deviation

$$\mu = E\{x|\omega\} \approx \frac{1}{N} \sum_{i=1}^N x_i \quad \text{with } x_i \in \omega, \quad N = |\omega|$$

$$\sigma^2 = E\{(x - \mu)^2|\omega\} \approx \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2$$

Stochastic Processes – Parameter Estimation

- Estimation of unimodal exponential distribution

- Necessary parameter: λ

$$p(x|\lambda) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

- Maximum likelihood (ML) estimator for exponential parameter

$$\frac{1}{\lambda} = E\{x|\omega\} \approx \frac{1}{N} \sum_{i=1}^N x_i \quad \text{with } x_i \in \omega, \quad N = |\omega|$$

- Estimation of multinomial distribution

- Count the frequency of occurrence of each element in the sample (→ histogram)
- Probabilities result from the frequencies

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Stochastic Processes – Multivariate Densities

► Generalization of densities to several random variables

- Multivariate distributions with multidimensional random variables

$$x \in \mathbb{R} \quad \Rightarrow \quad \mathbf{x} \in \mathbb{R}^n$$

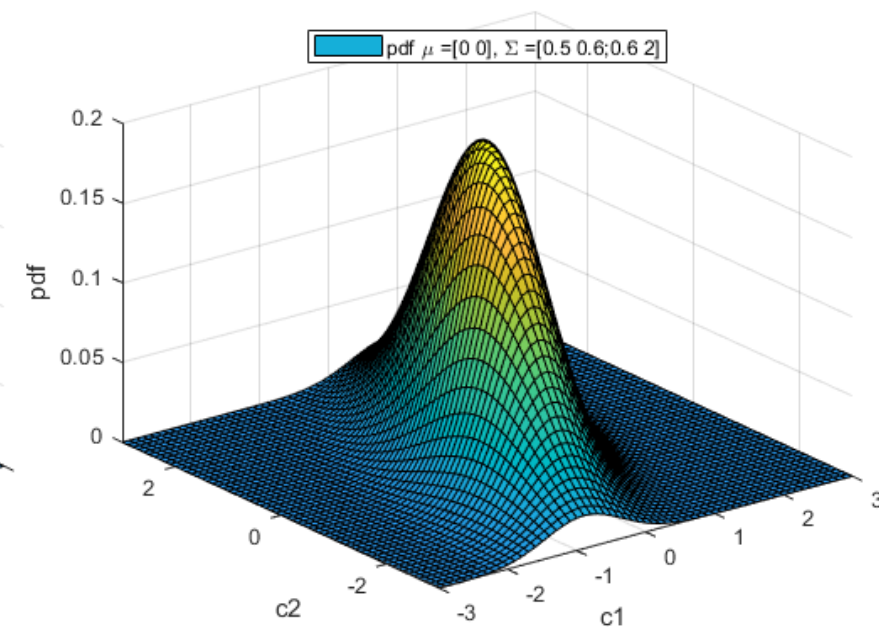
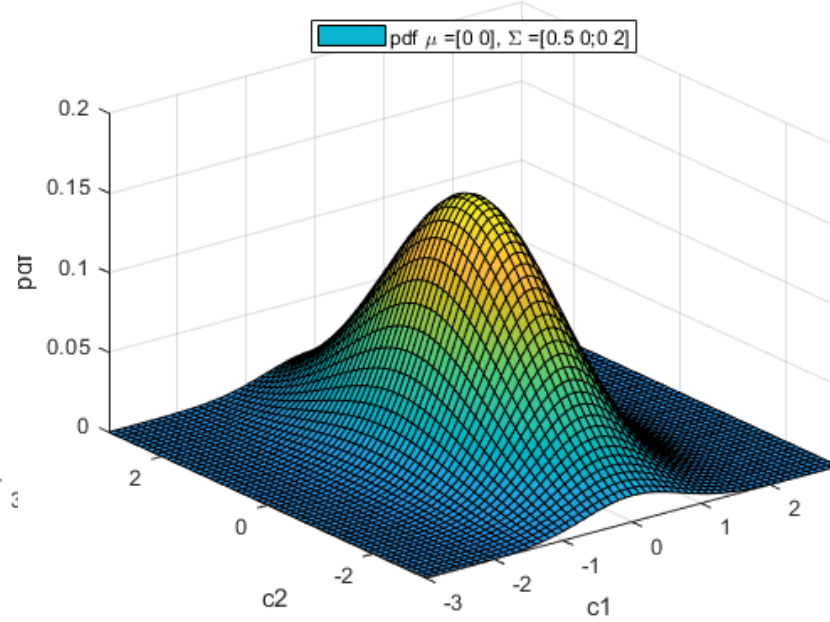
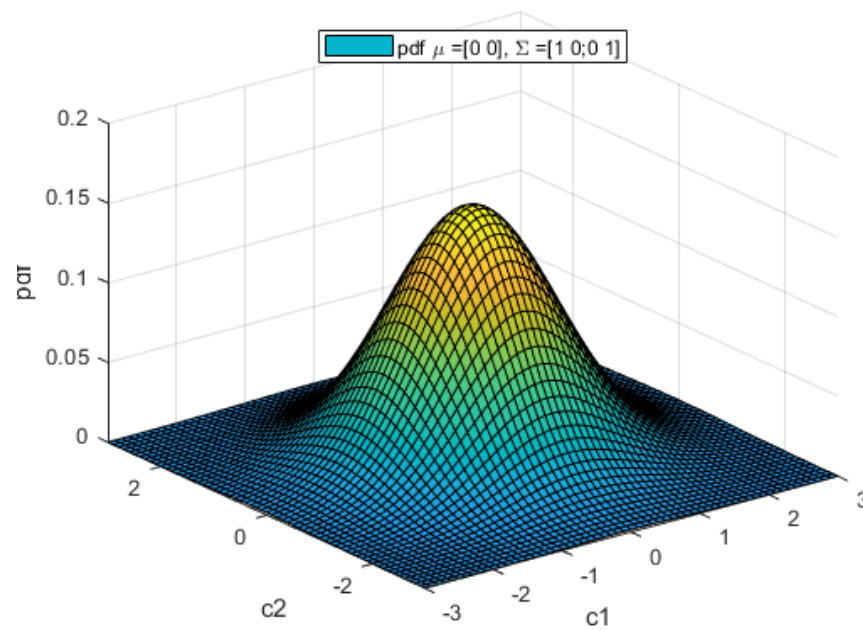
- Uniform density straightforward: Each dimension/feature has own upper and lower limits
- Very important: Multivariate normal distribution

$$p(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{\sqrt{\det(2\pi\boldsymbol{\Sigma})}} \exp\left(-\frac{(\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu})}{2}\right)$$

- Parameters: mean vector and covariance matrix
- Mean vector is the position where the distribution has maximum value
- Covariance matrix is a square matrix that “shapes” the normal distribution
 - Covariance between each pair of feature elements
 - Symmetric and positive semi-definite matrix
 - Main diagonal contains variances (squared standard deviations)

Stochastic Processes – Multivariate Densities

- Example plots for different covariance matrices



Stochastic Processes – Multivariate Densities

- Sample from arbitrary normally distributed density → Create random numbers that follow this distribution
 - Given the parameters of that distribution: mean μ and covariance matrix Σ
 - Create multivariate standard-normally distributed random numbers

$$x_1, x_2, \dots, x_n \sim \mathcal{N}(0.0, 1.0)$$

$$\mathbf{x} = (x_1, x_2, \dots, x_n)^T$$

$$\Rightarrow \mathbf{x} \sim \mathcal{N}(\mathbf{0}^{n \times 1}, \mathbf{I}^{n \times n})$$

- Use Cholesky decomposition of covariance matrix to transform random numbers

$$\Sigma = \mathbf{L}\mathbf{L}^T$$

$$\mathbf{z} = \mathbf{L}\mathbf{x} + \mu$$

$$\Rightarrow \mathbf{z} \sim \mathcal{N}(\mu, \Sigma)$$

Stochastic Processes – Multivariate Densities

- Estimation of arbitrary multivariate normal distribution
 - Calculate ML estimate from sample set

$$\boldsymbol{\mu} = E\{\boldsymbol{x}|\omega\} \approx \frac{1}{N} \sum_{i=1}^N \boldsymbol{x}_i \quad \text{with } \boldsymbol{x}_i \in \omega, \quad N = |\omega|$$

$$\boldsymbol{\Sigma} = E\{(\boldsymbol{x} - \boldsymbol{\mu})(\boldsymbol{x} - \boldsymbol{\mu})^T|\omega\} \approx \frac{1}{N} \sum_{i=1}^N (\boldsymbol{x}_i - \boldsymbol{\mu})(\boldsymbol{x}_i - \boldsymbol{\mu})^T$$

- Multidimensional exponential distributions are rarely used
 - Multivariate Gamma distribution is extension of exponential distribution

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Sampling Problems – Importance Sampling

► „Houston, we have a problem“

- Estimation of parameters always requires a sufficiently large sample set for stable results
 - Estimation of parameters is calculation of expectations
- What if you don't get enough samples from your distribution in the necessary range?
→ Solution: Importance Sampling
- Example:
 - Let's assume we have samples that follow a standard normal distribution

$$x \sim \mathcal{N}(0.0, 1.0) = p(x)$$

- We want to estimate the expected, counted number the samples that are smaller than a given value

$$f_{\theta}(x) = \begin{cases} 1 & x < \theta \\ 0 & \text{otherwise} \end{cases}$$

$$E(f_{\theta}(x)) = \frac{1}{N} \sum_x f_{\theta}(x)$$

Sampling Problems – Importance Sampling

- This function calculates an approximation of the cumulative normal distribution function

$$E(f_\theta(x)) \approx \int_{-\infty}^{\theta} \mathcal{N}(x|0.0, 1.0) \, dx$$

- Simple example for value of 0

$$E(f_0(x)) \approx \quad ?$$

- Difficult example: very few samples are smaller than -20

$$\int_{-\infty}^{-20} \mathcal{N}(x|0.0, 1.0) \, dx \approx 10^{-89}$$

$$E(f_{-20}(x)) \approx \quad ?$$

Sampling Problems – Importance Sampling

- Cause of the problem: No samples occur in the “important” range $[-\infty, -20]$ for which we calculate the expectation

$$E(f_{\theta}(x)) = \int f_{\theta}(x)p(x)dx \approx \frac{1}{N} \sum_x f_{\theta}(x)$$

- Solution: Importance Sampling

- Take samples from a **proposal distribution** that generates many samples in the important range $[-\infty, -20]$, e.g. from

$$y \sim \mathcal{U}(-30.0, 2.0) = q(y)$$

- Use these samples for calculation of the expectation and correct the wrong probability of occurrence of the samples

$$E(f_{\theta}(x)) = \int f_{\theta}(x)p(x)dx = \int f_{\theta}(y)\frac{p(y)}{q(y)}q(y)dy \approx \frac{1}{N} \sum_y f_{\theta}(y)\frac{p(y)}{q(y)}$$

Sampling Problems – Importance Sampling

- In our example this gives the following results for sample sets of size 1 million

$$x \sim \mathcal{N}(0.0, 1.0) = p(x)$$

$$y \sim \mathcal{U}(-30.0, 2.0) = q(y)$$

for the limit at 0

$$E(f_0(x)) \approx 0.5$$

$$E\left(f_0(y) \frac{p(y)}{q(y)}\right) \approx 0.5$$

and the limit at -20

$$E(f_{-20}(x)) = 0$$

$$E\left(f_{-20}(y) \frac{p(y)}{q(y)}\right) \approx 2.78 \cdot 10^{-89}$$

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Stochastic Processes – Gaussian Mixture Model

- Mixture Models: very powerful technique for flexible densities
 - Gaussian mixture model: density model where a number of normal distributions are combined

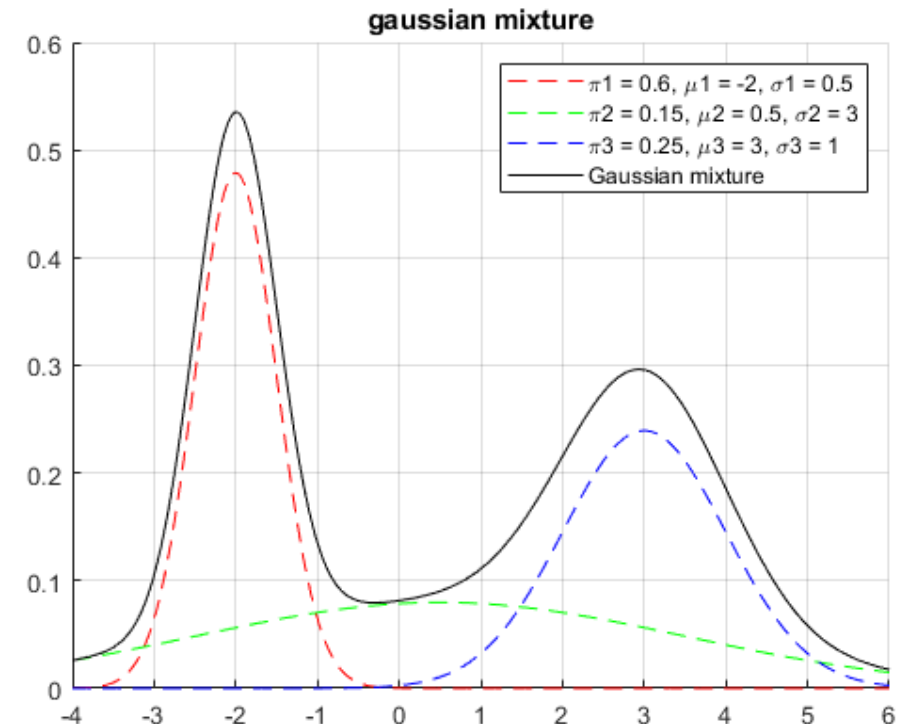
$$p(\mathbf{x}|\omega) = p(\mathbf{x}|\pi^1, \boldsymbol{\mu}^1, \boldsymbol{\Sigma}^1, \dots, \pi^J, \boldsymbol{\mu}^J, \boldsymbol{\Sigma}^J) = p(\mathbf{x}|\mathbf{a}) = \sum_{n=1}^J \pi^n \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}^n, \boldsymbol{\Sigma}^n)$$

j = 3 multi model problem covariance matrix

$$\mathbf{a} = \{\pi^1, \boldsymbol{\mu}^1, \boldsymbol{\Sigma}^1, \dots, \pi^J, \boldsymbol{\mu}^J, \boldsymbol{\Sigma}^J\}$$

$$\boldsymbol{\pi} = (\pi^1, \pi^2, \dots, \pi^J)^T, \quad 0 \leq \pi^n \leq 1, \quad \sum_{n=1}^J \pi^n = 1$$

- Example with 3 univariate mixture components
- Result is a multimodal density
- Multimodal approaches are the method of choice for difficult problems
- Assume for now that parameters are known



Stochastic Processes – Gaussian Mixture Model

- Draw sample from mixture density

- Step 1: Create discrete random number i that follows the multinomial distribution

$$i \sim M(\pi^1, \pi^2, \dots, \pi^J)$$

- Sampling one element from weights proportional to the weights gives probability of random number

$$p(i = n) = \pi^n$$

- How to do that?

- Step 2: Use i to sample from the corresponding mixture component

- Create random number that follows that single mixture component i

$$z \sim \mathcal{N}(\mu^i, \Sigma^i)$$

- Here for normal distribution, works similar for other underlying mixture components

Stochastic Processes – Gaussian Mixture Model

► Parameter estimation for obtaining a mixture model

- We still have a sample set

$$\omega, \quad \mathbf{x}_i \in \omega, \quad N = |\omega| \quad \mathbf{a} = \{\pi^1, \boldsymbol{\mu}^1, \boldsymbol{\Sigma}^1, \dots, \pi^J, \boldsymbol{\mu}^J, \boldsymbol{\Sigma}^J\}$$

- Likelihood of the sample set: the predictive distribution of the training data given the parameters

- Assumption: samples are independent and identically distributed
- This leads to the factorized likelihood

$$p(\omega|\mathbf{a}) = \prod_{i=1}^N p(\mathbf{x}_i|\mathbf{a}), \quad p(\mathbf{x}_i|\mathbf{a}) = \sum_{n=1}^J \pi^n \mathcal{N}(\mathbf{x}_i|\boldsymbol{\mu}^n, \boldsymbol{\Sigma}^n)$$

$$\begin{aligned} a \cdot b &= c \mid \log \\ \log(a \cdot b) &= \log(c) \\ \log a + \log b &= \log(c) \end{aligned}$$

- Every individual likelihood term is a Gaussian mixture density
- All further calculations use the log-likelihood

$$\log p(\omega|\mathbf{a}) = \sum_{i=1}^N \log p(\mathbf{x}_i|\mathbf{a}) = \underbrace{\sum_{i=1}^N \log \sum_{n=1}^J \pi^n \mathcal{N}(\mathbf{x}_i|\boldsymbol{\mu}^n, \boldsymbol{\Sigma}^n)}_{\text{log-likelihood}}$$

Stochastic Processes – Gaussian Mixture Model

- Our parameter estimation needs to find parameters that maximize the log-likelihood

$$\log p(\omega|\mathbf{a}) = \sum_{i=1}^N \log p(\mathbf{x}_i|\mathbf{a}) = \underbrace{\sum_{i=1}^N \log \sum_{n=1}^J \pi^n \mathcal{N}(\mathbf{x}_i|\boldsymbol{\mu}^n, \boldsymbol{\Sigma}^n)}_{\text{log-likelihood}}$$

Need to estimate these ML parameters

The diagram shows the equation for log-likelihood. Red circles highlight the parameters $\mathbf{a}$ in $p(\omega|\mathbf{a})$, the mixing coefficients π^n , and the Gaussian parameters $\boldsymbol{\mu}^n, \boldsymbol{\Sigma}^n$. Red arrows point from the text 'Need to estimate these ML parameters' to each of these circled terms. A bracket under the inner sum is labeled 'log-likelihood'.

- Bad news: there is no closed-form solution
- Good news: there is an iterative scheme to find good model parameters

$$\operatorname{argmax}_{\mathbf{a}} \log p(\omega|\mathbf{a})$$

→ EM algorithm it try to find parameter for maximum log likelihood

Stochastic Processes – Gaussian Mixture Model

► EM-Algorithm [Dem77] (nice explanation with examples in [Dei20])

1. Initialize all parameters π^n, μ^n, Σ^n

[deison hault book](#)

2. E-Step: Calculate the responsibilities

$$r^{in} = \frac{\pi^n \mathcal{N}(\mathbf{x}_i | \mu^n, \Sigma^n)}{\sum_{j=1}^J \pi^j \mathcal{N}(\mathbf{x}_i | \mu^j, \Sigma^j)} \quad R^n = \sum_{i=1}^N r^{in}$$

■ Contribution of the n^{th} mixture component for the i^{th} sample element

3. M-step: Re-estimate parameters using the current responsibilities

$$\mu^n = \frac{1}{R^n} \sum_{i=1}^N r^{in} \mathbf{x}_i \quad \Sigma^n = \frac{1}{R^n} \sum_{i=1}^N r^{in} (\mathbf{x}_i - \mu^n)(\mathbf{x}_i - \mu^n)^T \quad \pi^n = \frac{R^n}{N}$$

4. Iterate from (2.) until log-likelihood becomes stable

Stochastic Processes – Gaussian Mixture Model

- Remarks on the EM-Algorithm

- E-step

- Assign samples to components
 - Expected values of the weights are calculated ...
 - ... under the assumption that expected values of the incoming random variables (sample set) correspond to those calculated in the maximization step
 - This assumption is violated in the first step directly after initialization

- M-step

- Direct implementation of ML approach: Improve model so that it better fits the samples
 - Expected values of the probability distributions of the individual random variables (representing the sample set) are determined ...
 - ... for which the probability of the sample result occurring is maximized (→ This is ML)

- Advantages of EM-Algorithm

- Mathematical model of the data with normal distributions (EM can be applied to other distributions, we will ignore that)
 - Can detect and represent correlated components (→ using covariance matrix)

- Disadvantages of EM-Algorithm

- Finds only local optima → resulting model depends on initialization
 - How many components are optimal?

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Stochastic Processes – Similarity of Densities

► Given 2 densities: Is it possible to measure the similarity between them?

- Kullback–Leibler (KL) divergence (Kullback–Leibler distance, relative entropy)

- Kind of statistical distance

- Measures how much a model density q is different from a true density p

$$D_{\text{KL}}(p \parallel q) = E \left\{ \log \frac{p(x)}{q(x)} \right\} = \int p(x) \log \frac{p(x)}{q(x)} dx$$

- Interpretation: expected excess surprise from using q as model (instead of p) when actual distribution is p

- Distance measure, but not a metric

- Distance of density from itself must be zero

→ fulfilled?

- Distance between two densities must be always positive

→ fulfilled?

- Distance measure is symmetric

→ fulfilled?

- Triangle inequality must hold

→ fulfilled?

Stochastic Processes – Similarity of Densities

- Example: Distance between two normal distributions
 - There exists a closed form for KL distance in this case, where k denotes dimensionality

$$D_{\text{KL}}(\mathcal{N}_0 \parallel \mathcal{N}_1) = \frac{1}{2} \left(\text{tr}(\Sigma_1^{-1} \Sigma_0) + (\mu_1 - \mu_0)^T \Sigma_1^{-1} (\mu_1 - \mu_0) - k + \log \frac{\det(\Sigma_1)}{\det(\Sigma_0)} \right)$$

(proof: "Duchi - Derivations for Linear Algebra and Optimization", eLearning)

- Two normal distribution, one with moving mean

$\mathcal{N}_0$

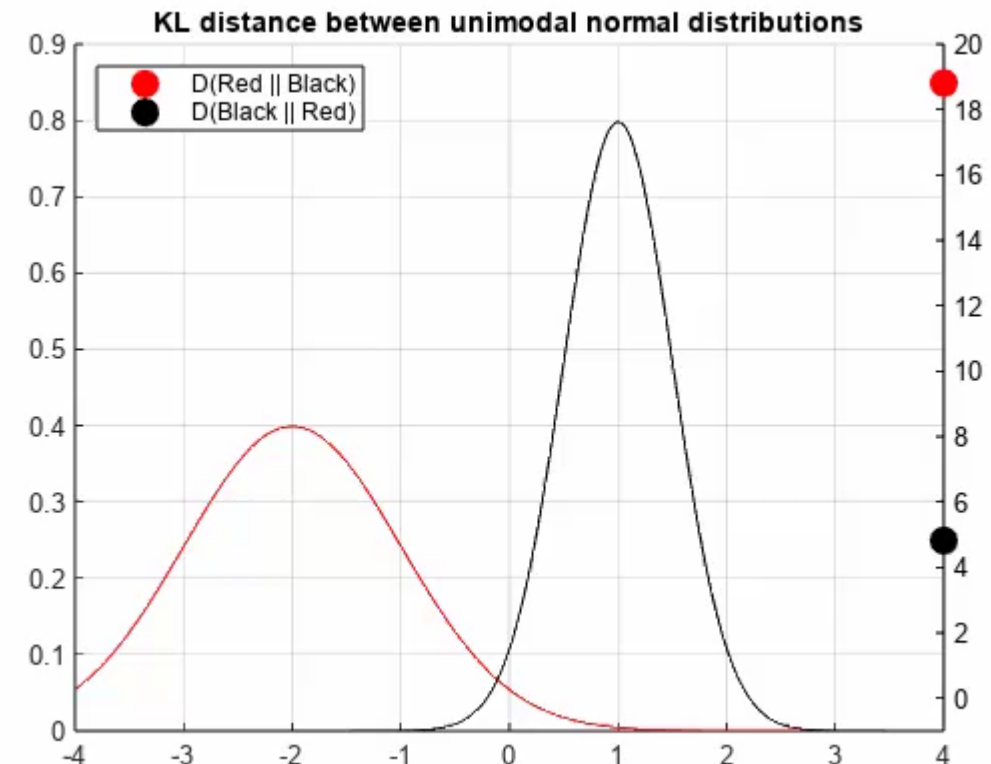
$\mathcal{N}_1$

- Calculating both KL distances

$D_{\text{KL}}(\mathcal{N}_0 \parallel \mathcal{N}_1)$

$D_{\text{KL}}(\mathcal{N}_1 \parallel \mathcal{N}_0)$

- Missing symmetry can be seen



Stochastic Processes – Similarity of Densities

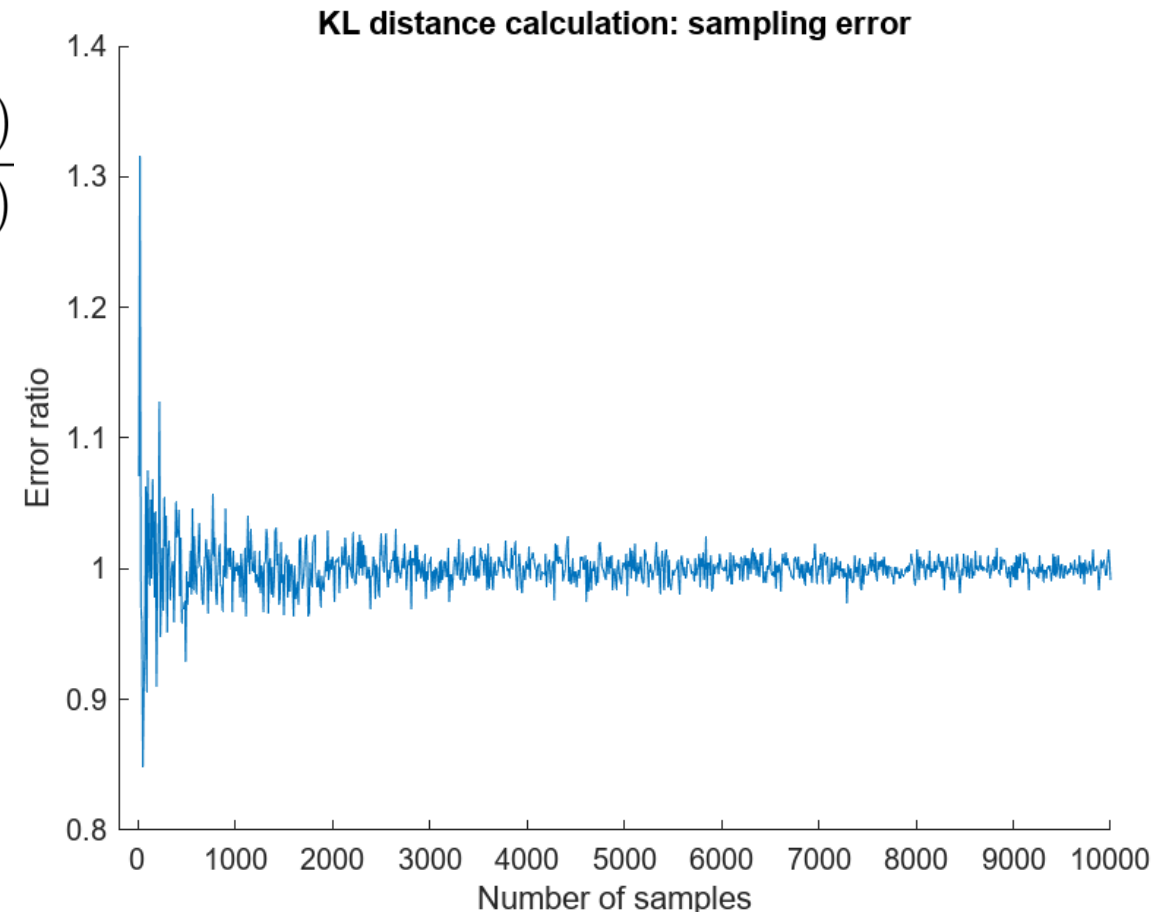
- Discrete Kullback-Leibler distance

- If there is no closed form for KL distance, then numerical calculation of integral is a bad idea
- KL distance can be approximated with discrete KL distance and based on sampling

$$D_{\text{KL}}(p \parallel q) = E \left\{ \log \frac{p(x)}{q(x)} \right\} = \sum_x p(x) \log \frac{p(x)}{q(x)}$$

- Example

- Same two normal distributions from last slide
 - Error of sampling compared to analytic solution
 - Converges quickly with number of samples
 - Numerically interesting for applications
-
- Should we expect numerical problems?



Stochastic Processes – Similarity of Densities

- Extended Kullback-Leibler distance

- If symmetric distance is required, the extended KL distance can be used

$$D_{\text{EKL}}(p \parallel q) = D_{\text{EKL}}(q \parallel p) = D_{\text{KL}}(p \parallel q) + D_{\text{KL}}(q \parallel p)$$

- Requires calculation of KL distance twice